



UNIVERSITY OF THE WESTERN CAPE
TUTORIAL 2
WEEK 3 (31 – 04 SEP)

**INFORMATION SYSTEMS
244**

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1 PL/SQL code

The screenshot shows the Oracle SQL Developer interface. The top menu bar includes Apple logo, Windows 11, File, Edit, View, Actions, Devices, Window, and Help. The main window has a title bar with 'Windows 11'. On the left, the 'Connections' pane shows a tree structure of database objects: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PC, MANAGER_ID, DEPARTMENT_ID, JOB_HISTORY, JOBS, JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY, and LOCATIONS. The 'Reports' pane is also visible. The main workspace is titled 'SQL Worksheet' and contains a 'Query Builder' tab. The code in the worksheet is as follows:

```
DECLARE
    sFirstname    VARCHAR2(20) DEFAULT 'Steven';
    sLastname     VARCHAR2(20) DEFAULT 'King';
    sPosition     VARCHAR2(20) DEFAULT 'President';
    sHireDate     DATE DEFAULT TO_DATE('17/JUN/03', 'DD/MON/RR');
    iEmpYears     NUMBER;

BEGIN
    iEmpyears := TRUNC(MONTHS_BETWEEN(SYSDATE, sHireDate)/12);
    DBMS_OUTPUT.PUT_LINE(sFirstname || ' ' ||
                        sLastname || ' is the ' ||
                        sPosition || ' of the the company. ');
    DBMS_OUTPUT.PUT_LINE('He was hired on ' ||
                        TO_CHAR(sHireDate, 'DD/MON/RR') ||
                        ' and has been employed for ' ||
                        iEmpYears || ' years. ');
END;
```

Output

The screenshot shows the Oracle SQL Developer interface with the 'Script Output' pane open. The 'Reports' pane on the left shows a tree structure of reports: All Reports, About Your Database, All Objects, Analytic View Reports, Application Express, ASH and AWR, Database Administration, and Data Dictionary. The 'Script Output' pane displays the following output:

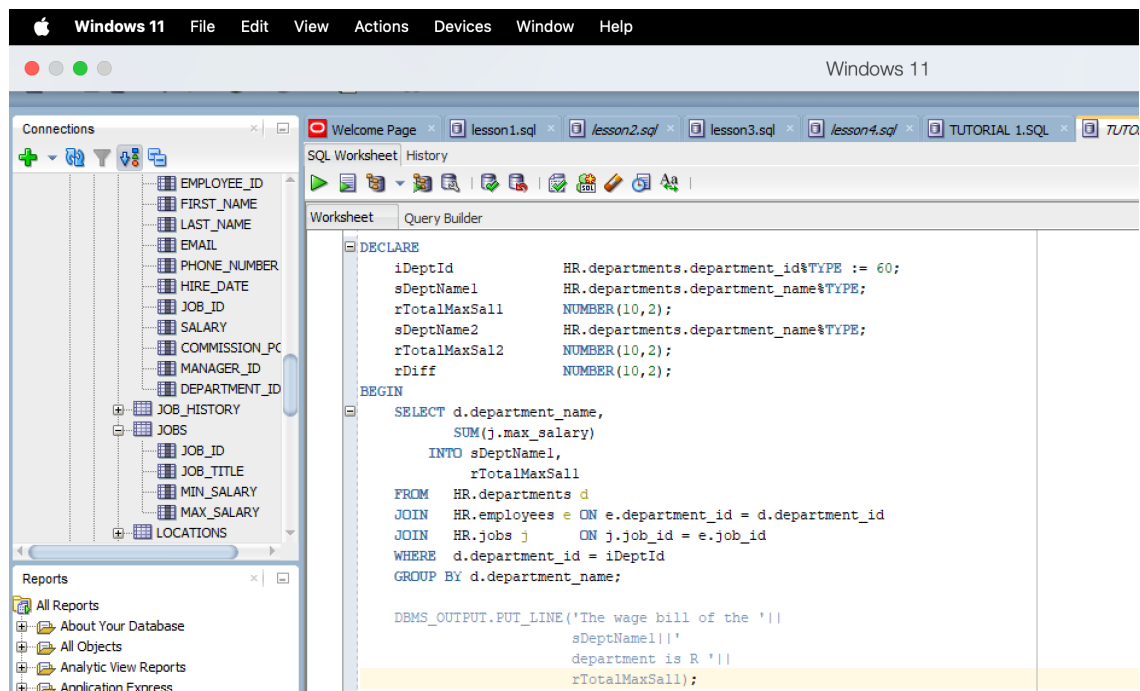
```
Task completed in 0.327 seconds

Steven King is the President of the the company.
He was hired on 17/JUN/03 and has been employed for 23 years.

PL/SQL procedure successfully completed.
```

2 PL/SQL code

Part 1

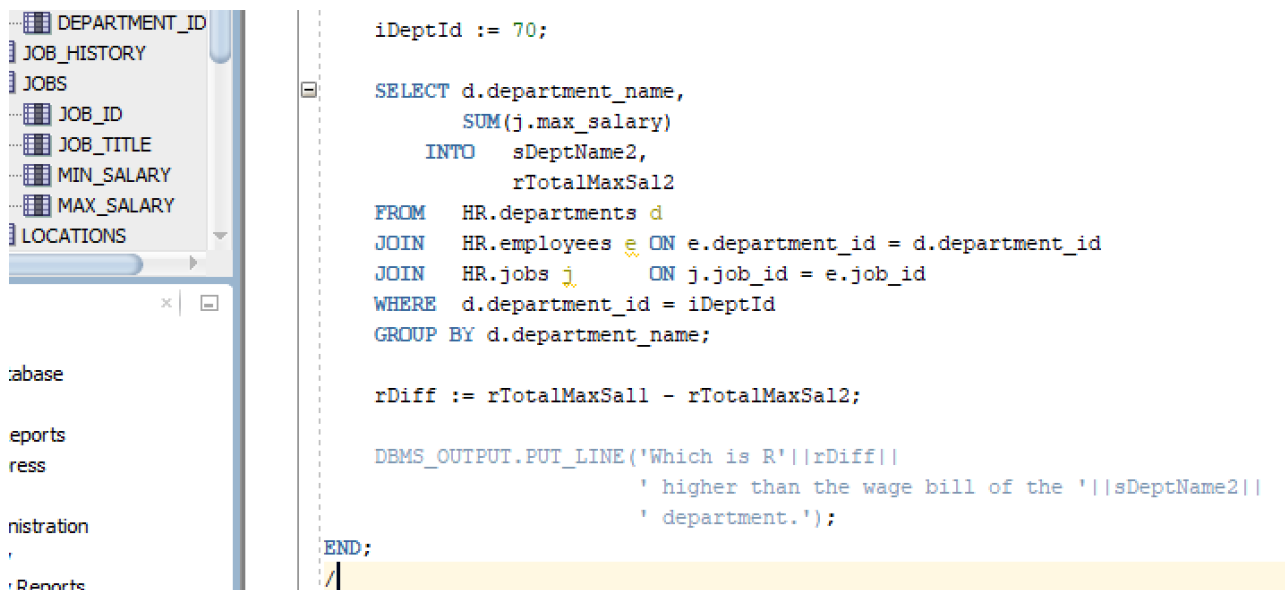


The screenshot shows the Oracle SQL Developer interface. On the left, the 'Connections' pane displays a tree view of the database schema, including tables like EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID, JOB_HISTORY, JOBS, MIN_SALARY, MAX_SALARY, and LOCATIONS. The main window displays a PL/SQL script in a worksheet. The script declares variables for department ID, department name, total max salary, and difference, then uses a SELECT statement to calculate the total max salary for a specific department (iDeptId = 60) and outputs the result using DBMS_OUTPUT.PUT_LINE.

```
DECLARE
iDeptId          HR.departments.department_id%TYPE := 60;
sDeptName1       HR.departments.department_name%TYPE;
rTotalMaxSal1    NUMBER(10,2);
sDeptName2       HR.departments.department_name%TYPE;
rTotalMaxSal2    NUMBER(10,2);
rDiff            NUMBER(10,2);
BEGIN
    SELECT d.department_name,
           SUM(j.max_salary)
           INTO sDeptName1,
               rTotalMaxSal1
    FROM   HR.departments d
    JOIN   HR.employees e ON e.department_id = d.department_id
    JOIN   HR.jobs j     ON j.job_id = e.job_id
    WHERE  d.department_id = iDeptId
    GROUP BY d.department_name;

    DBMS_OUTPUT.PUT_LINE('The wage bill of the '||
                          sDeptName1||
                          ' department is R '||
                          rTotalMaxSal1);
```

Part 2



The screenshot shows the Oracle SQL Developer interface. On the left, the 'Connections' pane displays a tree view of the database schema, including tables like DEPARTMENT_ID, JOB_HISTORY, JOBS, JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY, and LOCATIONS. The main window displays a PL/SQL script in a worksheet. The script declares variables for department ID, department name, total max salary, and difference, then uses a SELECT statement to calculate the total max salary for a specific department (iDeptId = 70) and outputs the result using DBMS_OUTPUT.PUT_LINE.

```
DECLARE
iDeptId := 70;

SELECT d.department_name,
       SUM(j.max_salary)
       INTO sDeptName2,
           rTotalMaxSal2
FROM   HR.departments d
JOIN   HR.employees e ON e.department_id = d.department_id
JOIN   HR.jobs j     ON j.job_id = e.job_id
WHERE  d.department_id = iDeptId
GROUP BY d.department_name;

rDiff := rTotalMaxSal1 - rTotalMaxSal2;

DBMS_OUTPUT.PUT_LINE('Which is R'||rDiff||
                     ' higher than the wage bill of the '||sDeptName2||
                     ' department.');
```

Output

```
END;  
/
```

Script Output

Task completed in 0.381 seconds

The wage bill of the IT department is R 50000
Which is R39500 higher than the wage bill of the Public Relations department.

PL/SQL procedure successfully completed.

